

Route quality: Voronoi vs PRM vs Visibility Graph

Visibility Graph reaches the goal fastest (3.22 s) and Voronoi slowest (3.56 s) at event @ 20 ms. One-shot routes are longer than the shortest path by a median of Voronoi 31 %, PRM 13 %, Visibility Graph 0 %. Finished runs only, same 200 scenarios per arm. Heading change = total turning per metre driven; sharp turn = a change of direction over 90°. Closest approach = smallest gap between robot and obstacle surfaces in a run. In each group, ▲ blue (bold) marks the best planner and ▼ orange the worst; marked only when they differ by the stated margin (3 % for time, 2 % for distance, 10 % otherwise).

	Time to goal, median (s)	Distance driven / straight line	Heading change (rad per m)	Sharp turns per run	Closest approach to an obstacle, median (mm)
Event (ours) @ 20 ms					
Voronoi	3.56 ▼	1.07	0.79 ▼	4.9	31 ▲
PRM	3.34	1.08 ▼	0.73	5.0 ▼	31
Visibility Graph	3.22 ▲	1.00 ▲	0.44 ▲	2.7 ▲	25 ▼
Event (ours) @ 20 ms, 50 ms horizon					
Voronoi	3.92 ▼	1.10	1.06 ▼	6.5	62
PRM	3.73 ▲	1.14 ▼	0.94	6.8 ▼	65 ▲
Visibility Graph	3.74	1.01 ▲	0.69 ▲	4.1 ▲	54 ▼
Cycle @ 20 ms					
Voronoi	3.24 ▼	1.07 ▼	1.51 ▼	9.7	31
PRM	2.99 ▲	1.06	1.08	7.2 ▲	34 ▲
Visibility Graph	3.12	1.00 ▲	1.07 ▲	10.6 ▼	22 ▼